Agentic Data Generation

1 Domain & Scenario



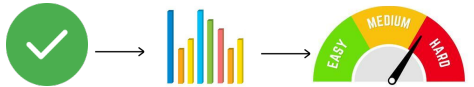
Enterprise Domains → Scenario Synthesis → Use case instantiation

2 DAG & Tool Generation



6 DAG topologies → Tool schemas → Execution schemas

3 Validation & Difficulty

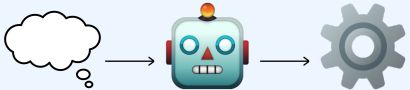


Schema validation → Programmatic Verification → 3 difficulty levels

O/P - 3,808 records with GT traces

Judge evaluation pipeline

a Tool call generation



User query + tools → Generator model → Generator tool calls

5 generators: SmolLM3-3B, Llama-3.1-8B, Qwen3-32B, Llama-3.3-70B, GPT-5.4

b Judge Scoring

With GT



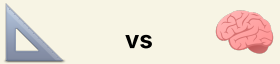
Expected tools

Without GT



6 judge configs: 20B → frontier scale

c Evaluating Judge



Programmatic verdict **vs** Judge model verdict

- Tool calling
- Parameter structure
- Sequence acc.
- Query coverage